Experiment 4

Aim: To implement Advanced Text Pre-processing Techniques

Code:

a. Stop word removal, lemmatization, stemming.

```
%pip install nltk
```

```
Requirement already satisfied: nltk in c:\users\tanu bhardwaj\appdata\local\programs\python\python313\lib\site-packages (3.10.3)
Requirement already satisfied: defusedxml in c:\users\tanu bhardwaj\appdata\local\programs\python\python313\lib\site-packages (from nltk) (0.7.1)
Requirement already satisfied: click in c:\users\tanu bhardwaj\appdata\local\programs\python\python313\lib\site-packages (from nltk) (8.4.2)
Requirement already satisfied: joblib in c:\users\tanu bhardwaj\appdata\local\programs\python\python313\lib\site-packages (from nltk) (1.5.3)
Requirement already satisfied: regex>=2021.8.3 in c:\users\tanu bhardwaj\appdata\local\programs\python\python313\lib\site-packages (from nltk) (2026.6.
28)
Requirement already satisfied: tqdm in c:\users\tanu bhardwaj\appdata\local\programs\python\python313\lib\site-packages (from nltk) (4.67.3)
Requirement already satisfied: colorama in c:\users\tanu bhardwaj\appdata\local\programs\python\python313\lib\site-packages (from click->nltk) (0.4.6)
Note: you may need to restart the kernel to use updated packages.


[notice] A new release of pip is available: 24.2 -> 26.2.1
[notice] To update, run: C:\Users\Tanu Bhardwaj\AppData\Local\Programs\Python\Python313\python.exe -m pip install --upgrade pip
```

```
import nltk

nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('stopwords')
nltk.download('wordnet')
nltk.download('omw-1.4')
```

```
[nltk_data] Downloading package punkt to C:\Users\Tanu
[nltk_data]     Bhardwaj\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package punkt_tab to C:\Users\Tanu
[nltk_data]     Bhardwaj\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
[nltk_data] Downloading package stopwords to C:\Users\Tanu
[nltk_data]     Bhardwaj\AppData\Roaming\nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to C:\Users\Tanu
[nltk_data]     Bhardwaj\AppData\Roaming\nltk_data...
[nltk_data] Downloading package omw-1.4 to C:\Users\Tanu
[nltk_data]     Bhardwaj\AppData\Roaming\nltk_data...
```

```python
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer, WordNetLemmatizer

text = "The students are studying and learning different programming concepts."

# Tokenization
words = word_tokenize(text.lower())

# Stopword removal
stop_words = set(stopwords.words('english'))
filtered_words = [word for word in words if word not in stop_words]

# Stemming
stemmer = PorterStemmer()
stemmed_words = [stemmer.stem(word) for word in filtered_words]

# Lemmatization
lemmatizer = WordNetLemmatizer()
lemmatized_words = [lemmatizer.lemmatize(word) for word in filtered_words]

print("Original Text:")
print(text)

print("\nAfter Stopword Removal:")
print(filtered_words)

print("\nAfter Stemming:")
print(stemmed_words)

print("\nAfter Lemmatization:")
print(lemmatized_words)
```

Output:

```
Original Text:
The students are studying and learning different programming concepts.

After Stopword Removal:
['students', 'studying', 'learning', 'different', 'programming', 'concepts', '.']

After Stemming:
['student', 'studi', 'learn', 'differ', 'program', 'concept', '.']

After Lemmatization:
['student', 'studying', 'learning', 'different', 'programming', 'concept', '.']
```

b. Handling noise in text data, multilingual text processing, and entity

anonymization.

```
import re

text = """
Hello!!! My name is Rahul Sharma.
My email is rahul@gmail.com and my phone number is 9876543210.
I am learning NLP. नमस्ते, मैं NLP सीख रहा हूँ।
"""

print("Original Text:")
print(text)

# 1. Noise removal
clean_text = re.sub(r'[!@#$%^&*(),.?":{}|<>]', '', text)
clean_text = re.sub(r'\s+', ' ', clean_text).strip()

print("\nAfter Noise Removal:")
print(clean_text)

# 2. Multilingual text processing
print("\nMultilingual Text:")
for word in clean_text.split():
    if re.search(r'[\u0900-\u097F]', word):
        print(word, "-> Hindi")
    elif re.search(r'[A-Za-z]', word):
        print(word, "-> English")

# 3. Entity anonymization
# Email anonymization
anonymized_text = re.sub(
    r'\b[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}\b',
    '[EMAIL]',
    clean_text
)
```

```
# Phone number anonymization
anonymized_text = re.sub(
    r'\b\d{10}\b',
    '[PHONE]',
    anonymized_text
)

# Name anonymization
anonymized_text = anonymized_text.replace(
    "Rahul Sharma",
    "[PERSON]"
)

print("\nAfter Entity Anonymization:")
print(anonymized_text)
```

Output:

```
Original Text:

Hello!!! My name is Rahul Sharma.
My email is rahul@gmail.com and my phone number is 9876543210.
I am learning NLP. नमस्ते, मैं NLP सीख रहा हूँ।


After Noise Removal:
Hello My name is Rahul Sharma My email is rahulgmailcom and my phone number is 9876543210 I am learning NLP नमस्ते मैं NLP सीख रहा हूँ।

Multilingual Text:
Hello -> English
My -> English
name -> English
is -> English
Rahul -> English
Sharma -> English
My -> English
email -> English
is -> English
rahulgmailcom -> English
and -> English
my -> English
phone -> English
number -> English
is -> English
I -> English
am -> English
learning -> English
NLP -> English
नमस्ते -> Hindi
मैं -> Hindi
NLP -> English
सीख -> Hindi
रहा -> Hindi
हूँ। -> Hindi


After Entity Anonymization:
Hello My name is [PERSON] My email is rahulgmailcom and my phone number is [PHONE] I am learning NLP नमस्ते मैं NLP सीख रहा हूँ।
```